Towards Naturally Fault-Tolerance



Rhapsody on a Theme by Kitaev



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1941 – First programmable electronic calculator.
mechanical relays.
1943 – ENIAC.
18000 vacuum tubes: “nature abhors the vacuum tube.”
1947 – Bell labs develops the transistor.
1952 – G. W. Dummer proposes manufacturing electronic equipment in one block with no connecting wires.
1959 – Texas Instruments and Fairchild Semiconductor invent the integrated circuit.
1964 – first IBM 360 series (the concept of an “architecture”)
1964 – Integrated circuit which cost $1000 in 1959 now costs $10. Moore describes his law.
1972 – Intel 8008 and 8 bit microprocessor.
1975 – First personal computer the Altair.
1982 – IBM PC introduced. Personal computer revolution begins.

Have We Learned Anything?

The Physics Guarantee

What is the phase of matter corresponding to the computer?

PRACTICAL QUESTION (as opposed to philosophical)

There are distinct PHYSICAL reasons why robust classical computation is possible.

not all physical systems are equally good for computation:

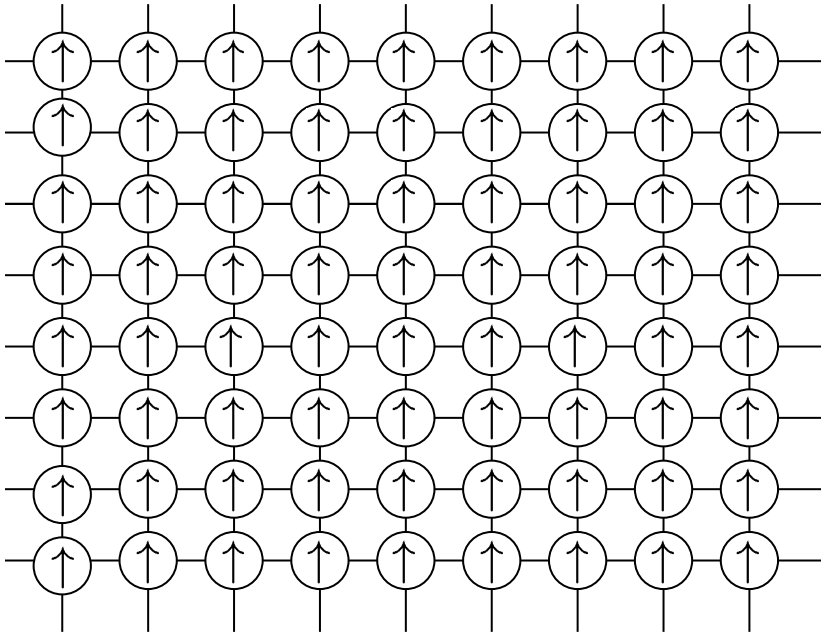
there exists systems whose PHYSICS guarantees their ability to enact robust classical computation.

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

This poster: (1) this point of view for classical computers

(2) an attempt to port these ideas to a quantum computer

A Dynamic Ising Model



Ising model

spins on a lattice $s_i \in \{+1, -1\}$

$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i$$

Energetically favorable to align

Order parameter:

$$M = \sum_i s_i$$

Dynamic Ising model (Metropolis update)

interaction with environment causes spins to flip

for every spin

if flipping spin would decrease energy, flip spin

if flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

Metropolis Monte Carlo

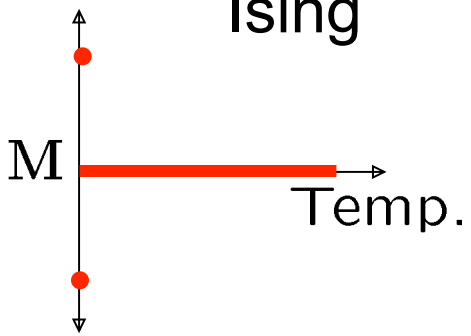
Storage and Manipulation

Compare 1D and 2D Ising models at different temperatures for
a) storage of information
b) manipulation of information

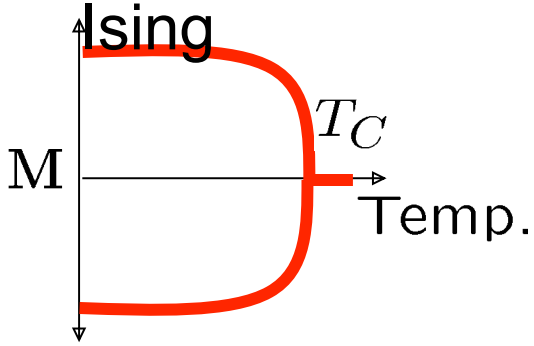
1st attempt:

Storage in thermal state (infinite time - ensemble is thermal)

1D Ising



2D Ising



Criticism: (1) thermodynamic limit taken (N infinite)
(2) what if relaxation to equilibrium takes a long time?

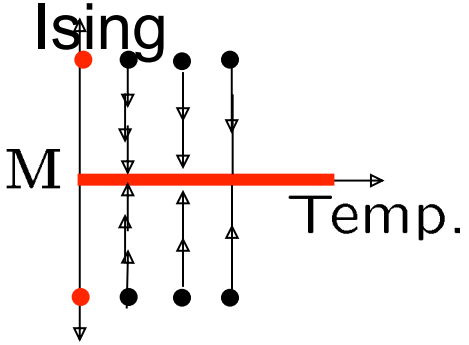
Storag



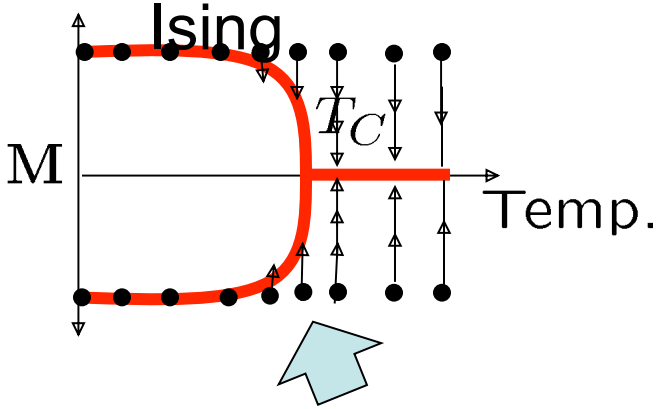
2nd attempt:

Relaxation to thermal equilibrium

1D Ising



2D Ising

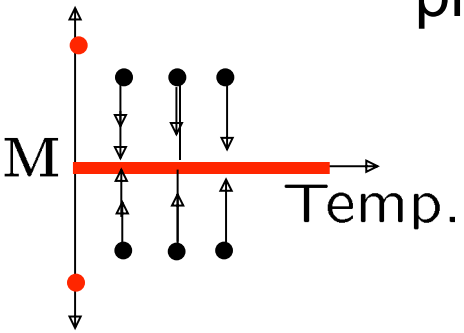


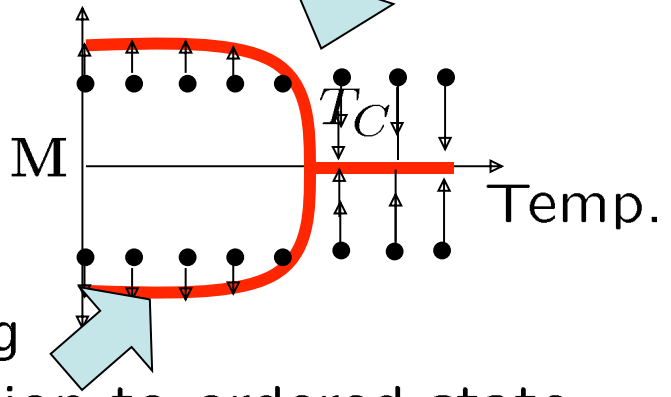
critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

decay rate $\propto e^{-\beta\Delta}$

Imperfect preparation





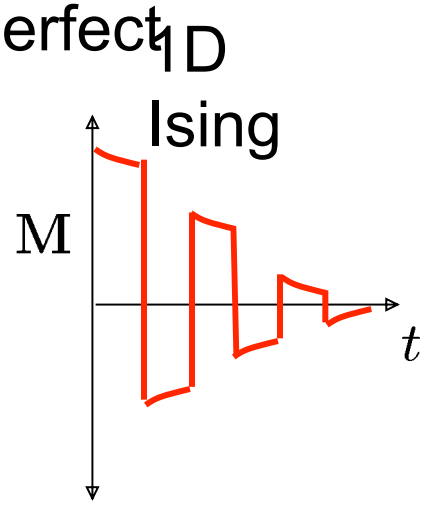
self correcting

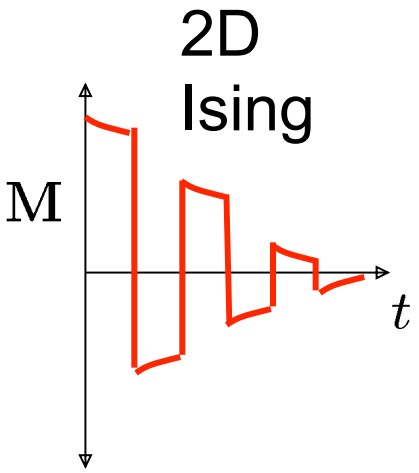
fast relaxation to ordered state

Manipulation



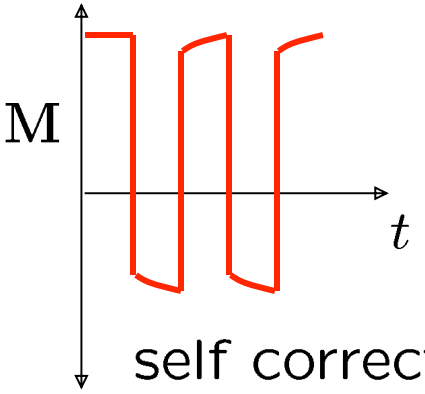
Flipping spins is imperfect





$T > T_C$

1D Ising
- temperature less than gap
- manipulation error must be small

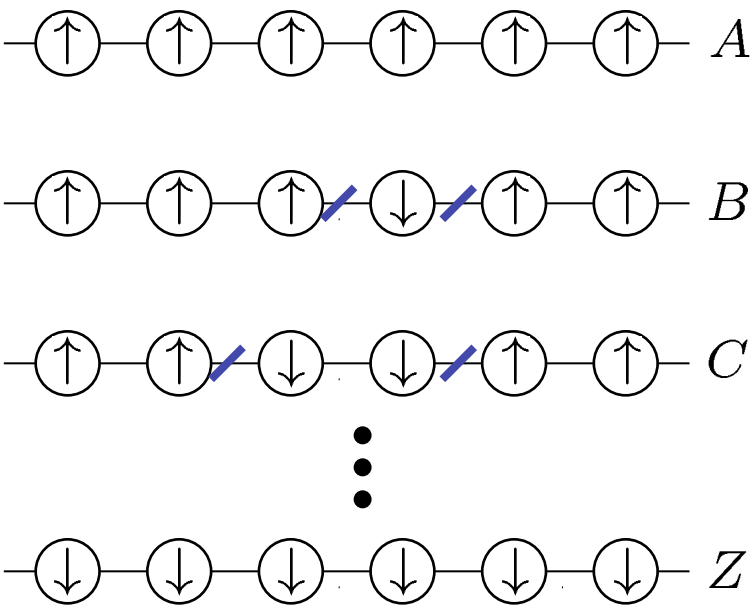


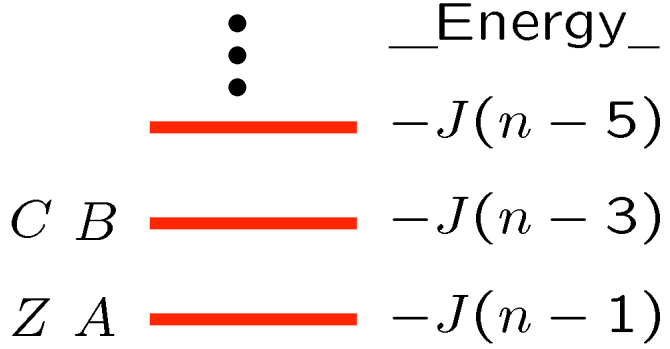
$T < T_C$

self correcting

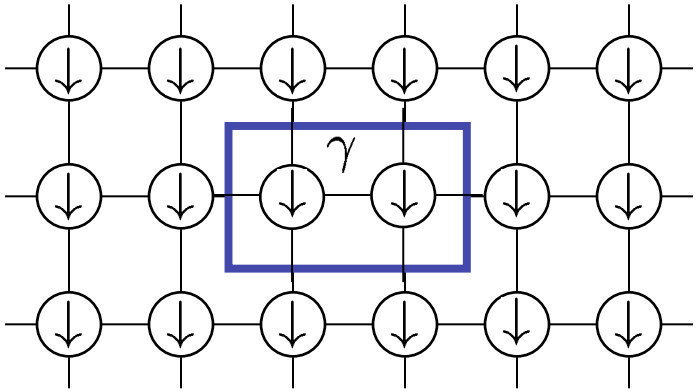
2D Ising
- temperature less than critical temp
- self correcting

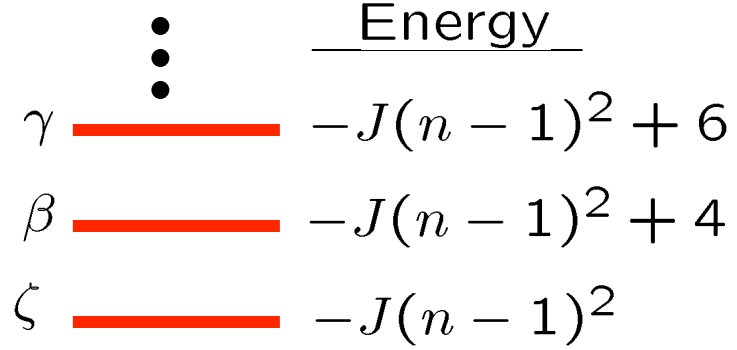
Self Fixing





Excitations free pointlike.
Energy cost constant.
disordering entropy $\gg$ energy



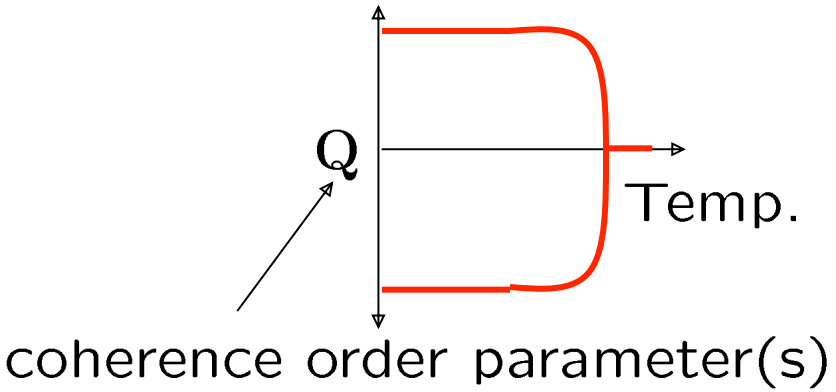


Excitations stringlike.
Energy cost grows like perimeter.
energy $\gg$ disordering entropy

The Quantum Hard Drive?

(Kitaev)

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



1. Coherence preserving.

$Q_1 \Rightarrow -Q_1$

macro local operations

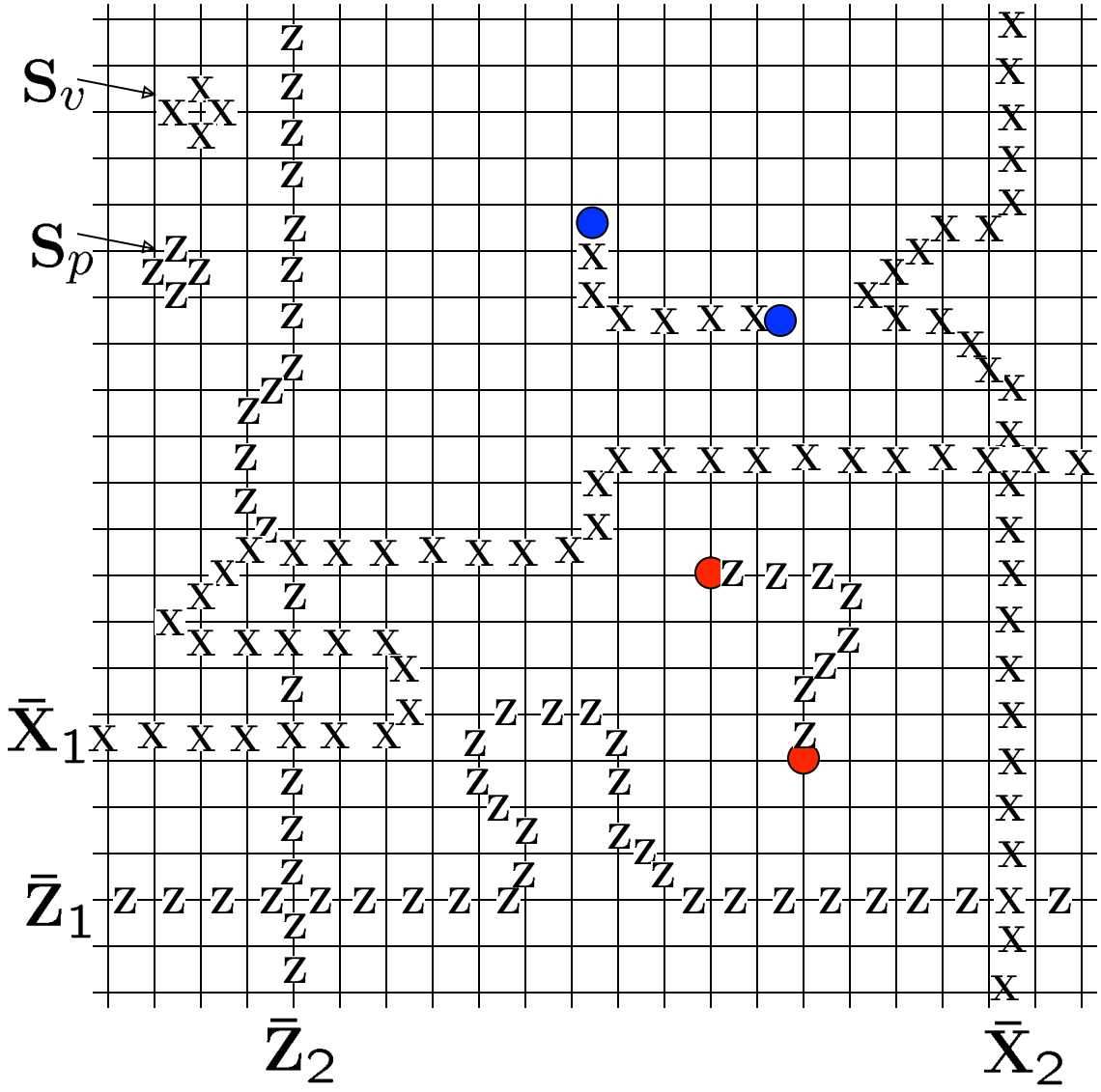
2. Accessible Fault-Tolerant Operations

"self-correcting"

$\langle G_1, \ldots, G_{n^k} \rangle \cong U(c^n)$

3. Universality

Wherefore Art Toric Codes?



$$\mathbf{S}_p = \sum_{e \in p} \mathbf{Z}_e$$
$$\mathbf{S}_v = \sum_{e \in v} \mathbf{X}_e$$

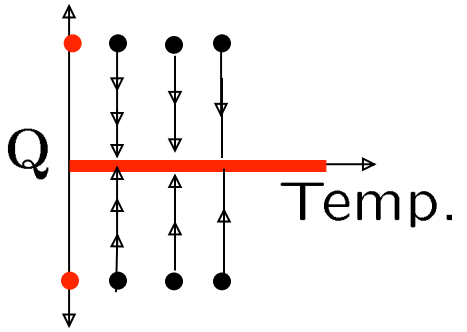
$$\mathbf{H} = -\frac{\Delta}{2}\left(\sum_p \mathbf{S}_p + \sum_v \mathbf{S}_v\right)$$

energy gap $= \Delta$

can correct almost any $O(L^2)$ qubit errors (percolation)

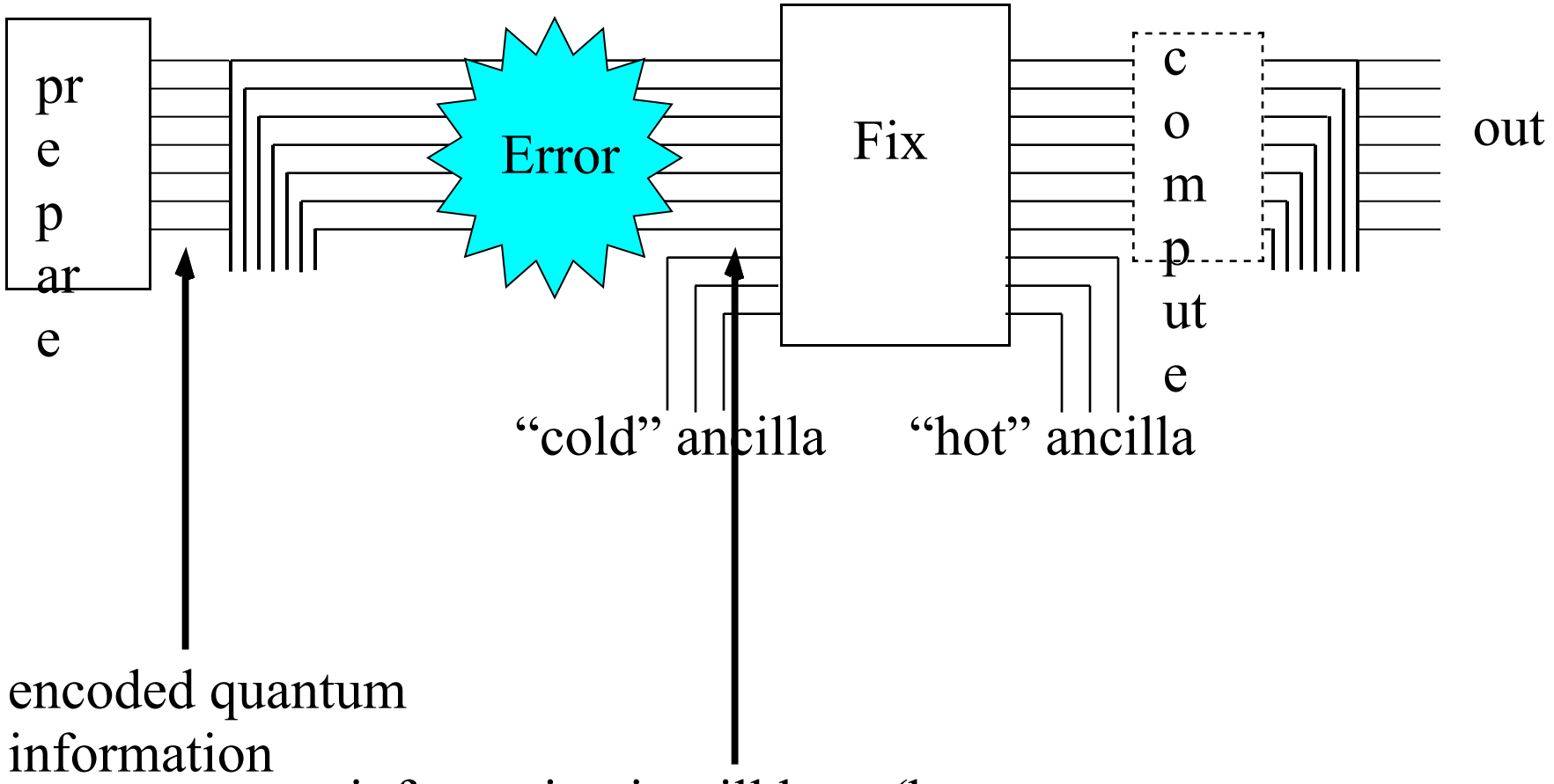
natural evolution is not self correcting

(we have to work in 2D)



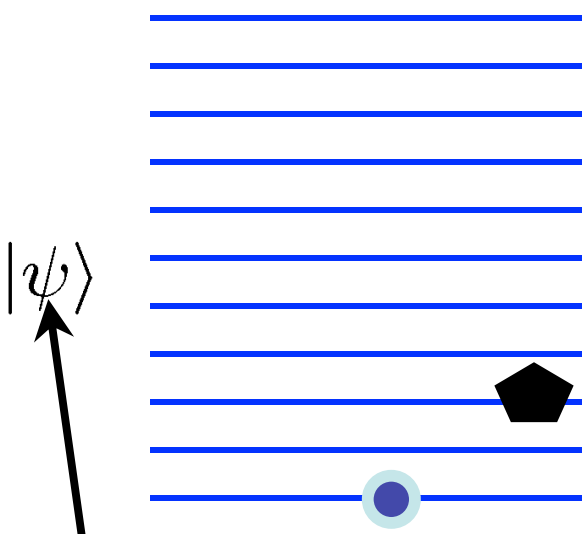
4D Toric code should be self correcting!

Old School



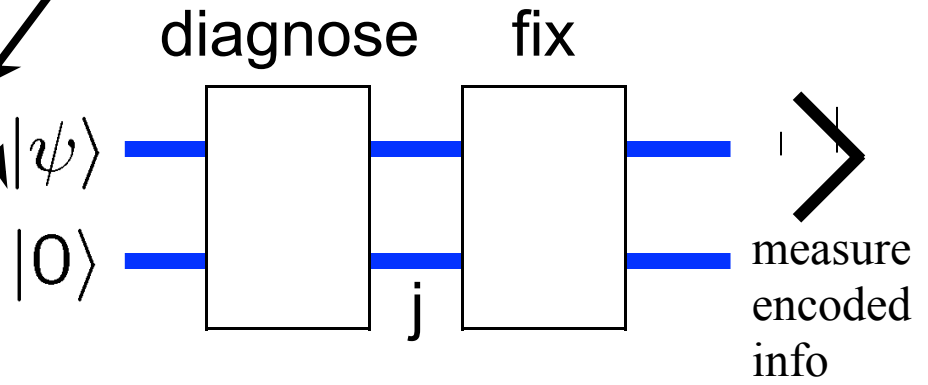
How do we reveal erred quantum information?

Quantum Error Correcting Order Parameter



ρ_{error}

The quantum information is still here. How do we see it?



Encoded quantum information

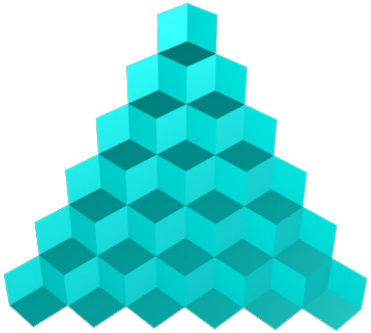
$$|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

$$\mathbf{Q}_s = \sum_j \mathbf{L}_s \mathbf{F}_j \mathbf{D}_j$$

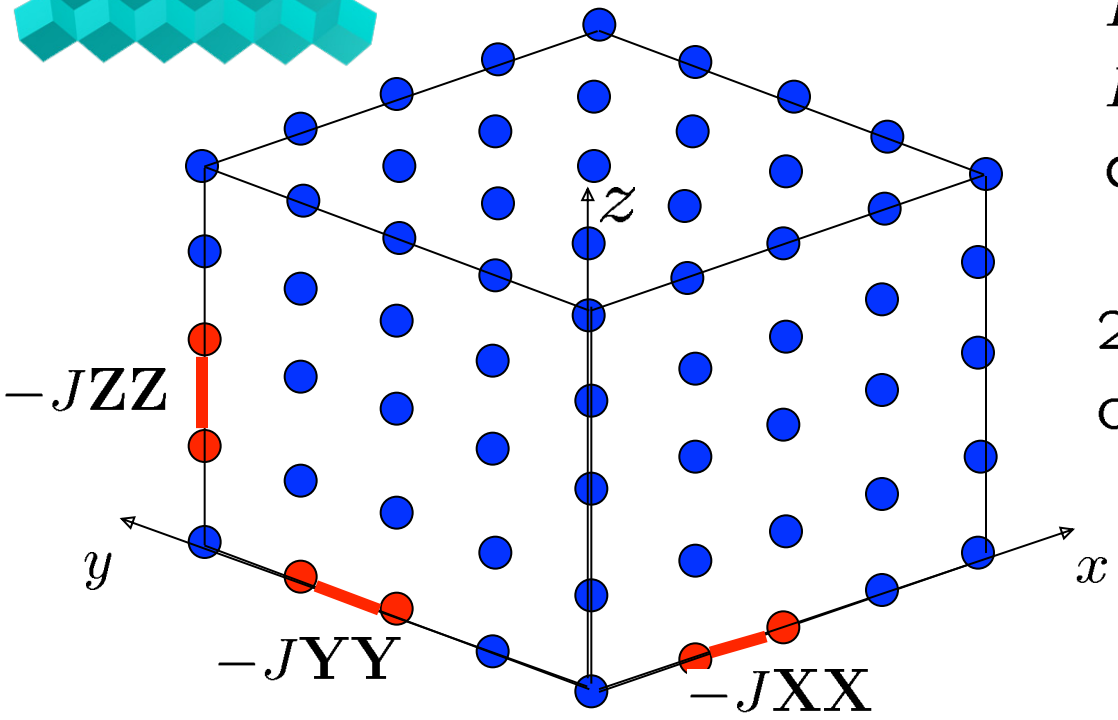
$\mathbf{L}_s$ information complete logical projectors

$\mathbf{D}_j$ projector onto jth errored subspace

$\mathbf{F}_j$ fixing operator for jth errored subspace



Into the Third Dimension



$L \times L \times L$ cubic lattice

L odd

qubits on vertices

2 qubit interactions along cartesian directions

An interesting Hamiltonian:

$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

Fudgy arguments about why persistent coherence…

How to proceed: simulate! But this systems has a “sign” problem.

Quantum Monte Carlo

Thermal ensemble:

$$\boldsymbol{\rho}_{th} = \frac{e^{-\beta \mathbf{H}}}{Z}, \quad Z = \mathrm{Tr}\left[e^{-\beta \mathbf{H}}\right]$$

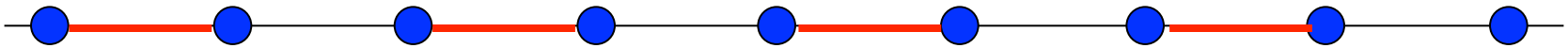
We want to understand coherence order parameters

$$\langle \mathbf{Q}_s \rangle = \mathrm{Tr}\left[\boldsymbol{\rho}_{th} \mathbf{Q}_s\right]$$

Since diagonalizing $\mathbf{H}$ is hard, how do we perform monte carlo?

Example: anisotropic Heisenberg chain

$$\mathbf{H}_{ex} = \sum_{i=1}^{n} \left(J_x \mathbf{X}_i \mathbf{X}_{i+1} + J_y \mathbf{Y}_i \mathbf{Y}_{i+1} + J_z \mathbf{Z}_i \mathbf{Z}_{i+1} \right)$$



$$\mathbf{H}_{ex} = \mathbf{H}_o + \mathbf{H}_e$$

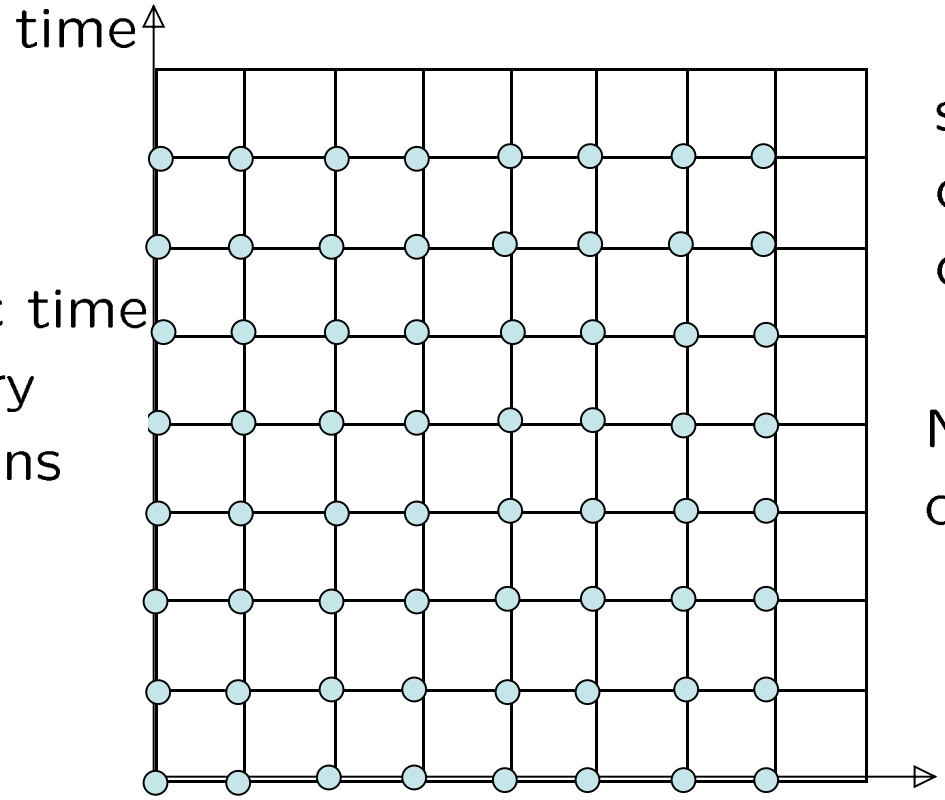
$$\mathbf{H}_o = \sum_{i=1|i \text{ odd}}^{n} \left(J_x \mathbf{X}_i \mathbf{X}_{i+1} + J_y \mathbf{Y}_i \mathbf{Y}_{i+1} + J_z \mathbf{Z}_i \mathbf{Z}_{i+1} \right)$$

$$\mathbf{H}_e = \sum_{i=1|i \text{ even}}^{n} \left(J_x \mathbf{X}_i \mathbf{X}_{i+1} + J_y \mathbf{Y}_i \mathbf{Y}_{i+1} + J_z \mathbf{Z}_i \mathbf{Z}_{i+1} \right)$$

Note fixed i terms commute for $\mathbf{H}_o$ and $\mathbf{H}_e$

Quantum Monte Carlo II

$$Z = \lim_{N\to\infty} \sum_{\{|s_k\rangle\}} \prod_{k=1,k \text{ odd}}^{2N} \langle s_k|e^{-\frac{\beta}{N}\mathbf{H}_o}|s_{k+1}\rangle\langle s_{k+1}|e^{-\frac{\beta}{N}\mathbf{H}_e}|s_{k+1}\rangle$$



Periodic time boudnary conditions

sum over classical configurations of $d+1$ dimensional lattice

Nonzero contribution only on colored plaquettes

Partition function on d dim quantum system

Partition-like function on $d+1$ dim classical system

Partition-Like = Sign Problem

$$\begin{bmatrix} e^{-\epsilon J_z}\cosh(\epsilon J_+) & 0 & 0 & e^{-\epsilon J_z}\sinh(\epsilon J_+) \\ 0 & e^{\epsilon J_z}\cosh(\epsilon J_-) & e^{\epsilon J_z}\sinh(\epsilon J_-) & 0 \\ 0 & e^{\epsilon J_z}\sinh(\epsilon J_-) & e^{\epsilon J_z}\cosh(\epsilon J_-) & 0 \\ e^{-\epsilon J_z}\sinh(\epsilon J_+) & 0 & 0 & e^{-\epsilon J_z}\cosh(\epsilon J_+) \end{bmatrix}$$

If J_- or J_+ is negative terms can be negative.

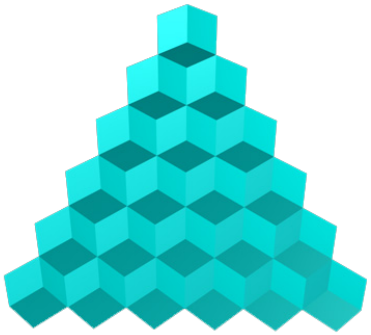
Partition function $= \sum$ positive terms $\Rightarrow$ probability interp

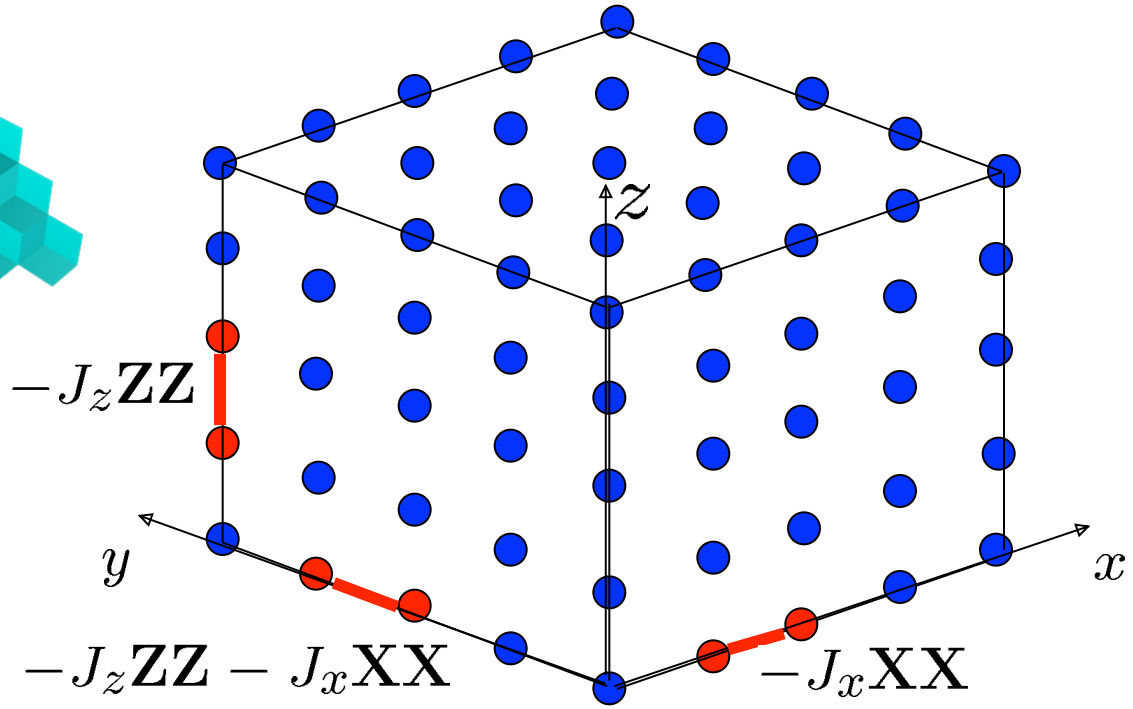
If there are negative terms, can't treat as probabilities!

SIGN PROBLEM

Sometimes sign problem can be fixed by basis change.

The Third Dimension Take II





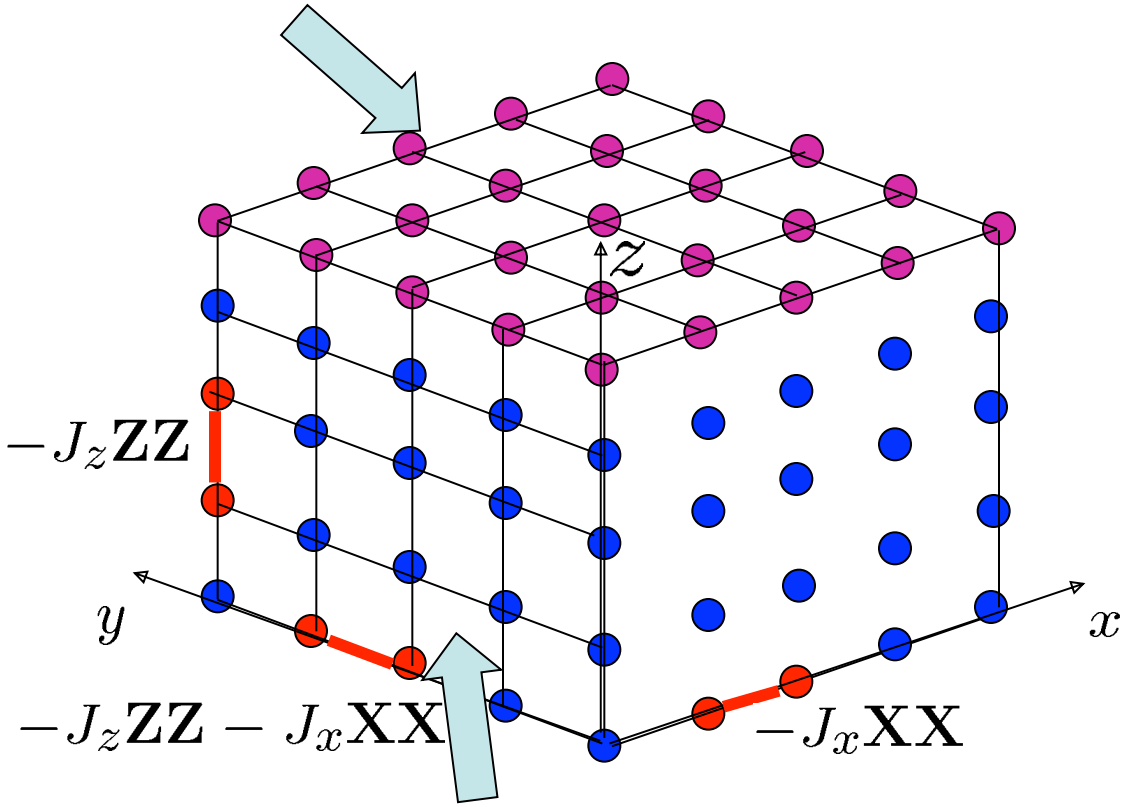
$$\mathbf{H} = -\sum_{ijk} J_x(\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{X}_{i,j,k}\mathbf{X}_{i,j+1,k})$$
$$- \sum_{ijk} J_z(\mathbf{Z}_{i,j,k} + \mathbf{Z}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1}))$$

No sign problem.

Stacks of Ising

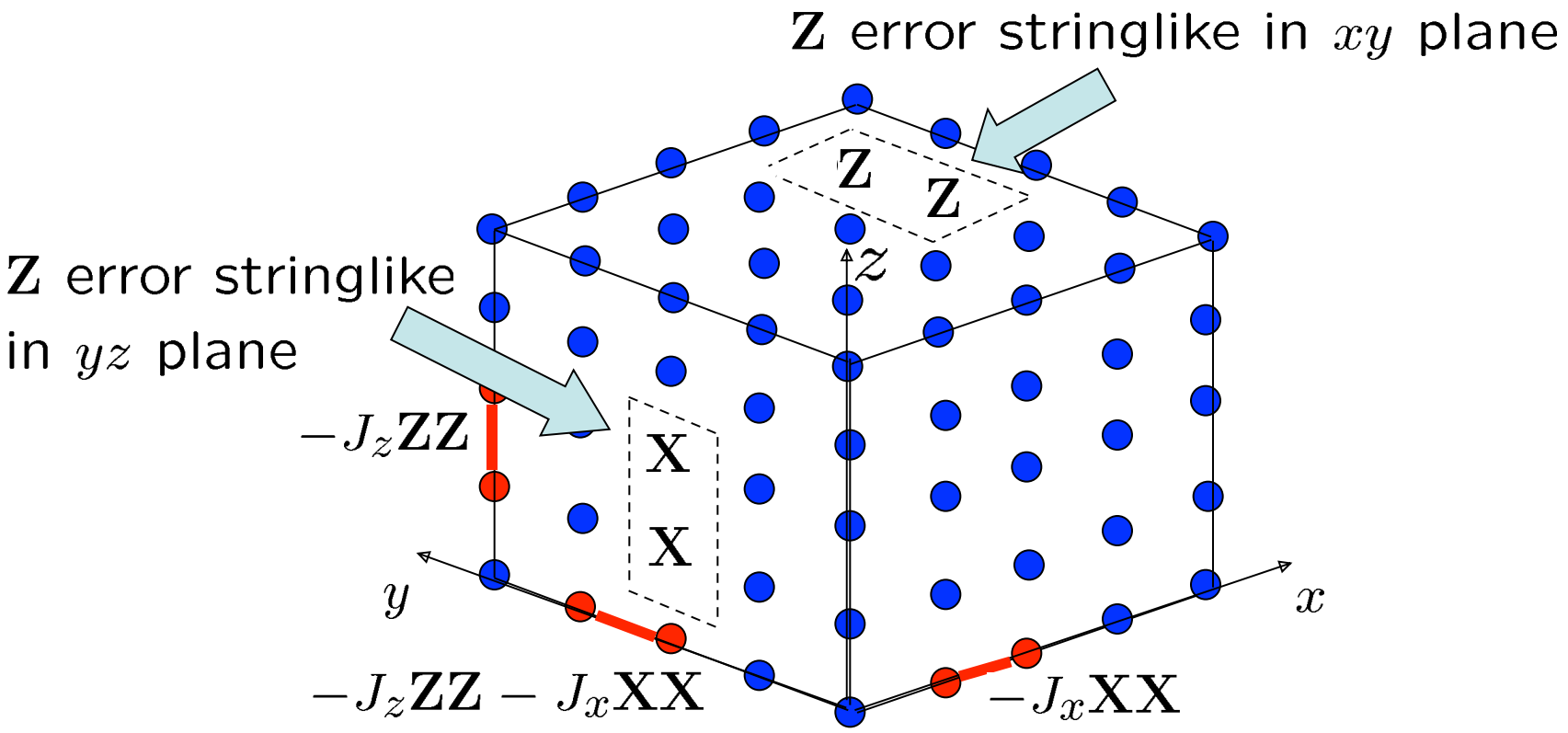


xy planes are 2D XX ising



yz planes are 2D ZZ ising

The Fudge for Stacks of Ising



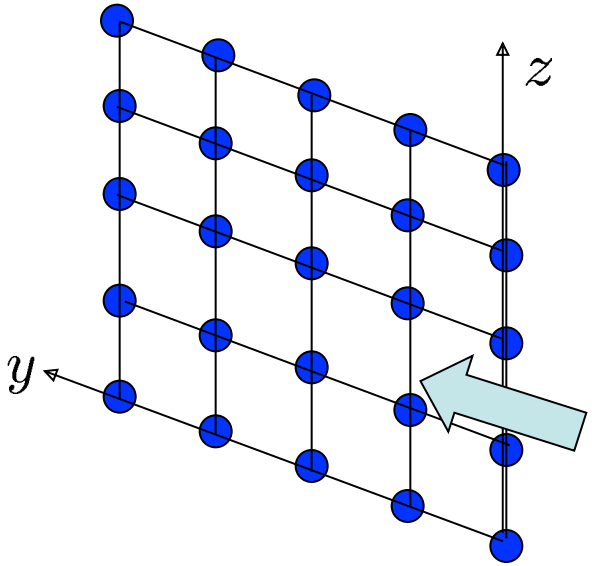
$\{\mathbf{XX}, \mathbf{ZI}\} = 0$, etc.

Group of Bonds (Barry and Bobby)

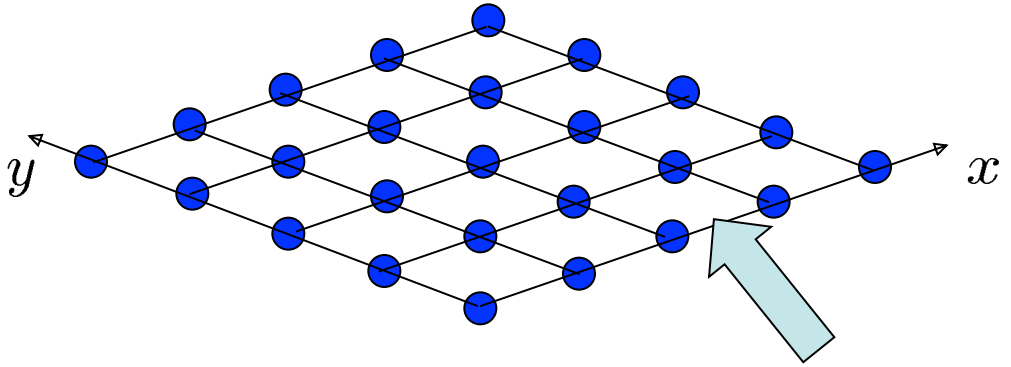
$$\begin{aligned}\mathbf{H} &= -\sum_{ijk} J_x(\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{X}_{i,j,k}\mathbf{X}_{i,j+1,k}) \\ &- \sum_{ijk} J_z(\mathbf{Z}_{i,j,k} + \mathbf{Z}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1}))\end{aligned}$$

Examine the group generated by the bonds

$$\mathcal{A} = \langle \mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k}, \mathbf{X}_{i,j,k}\mathbf{X}_{i,j+1,k}, \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j+1,k}, \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1} \rangle$$

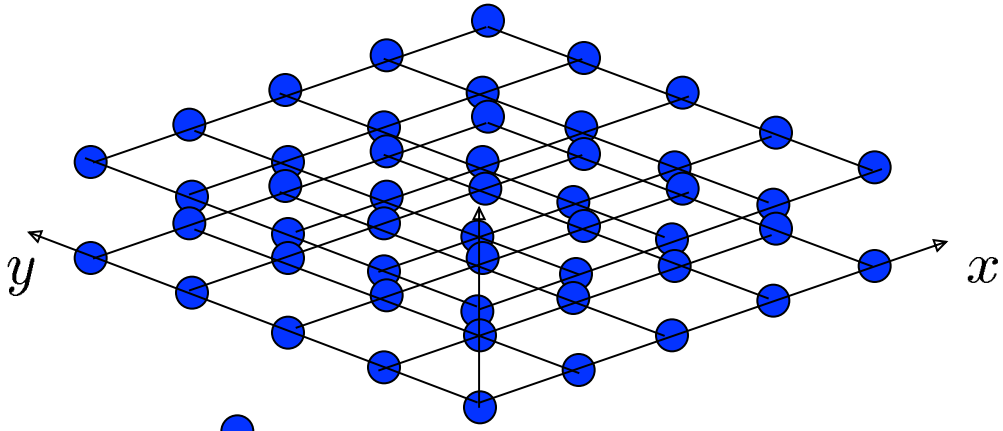


Even number of $\mathbf{Z}$s in yz planes



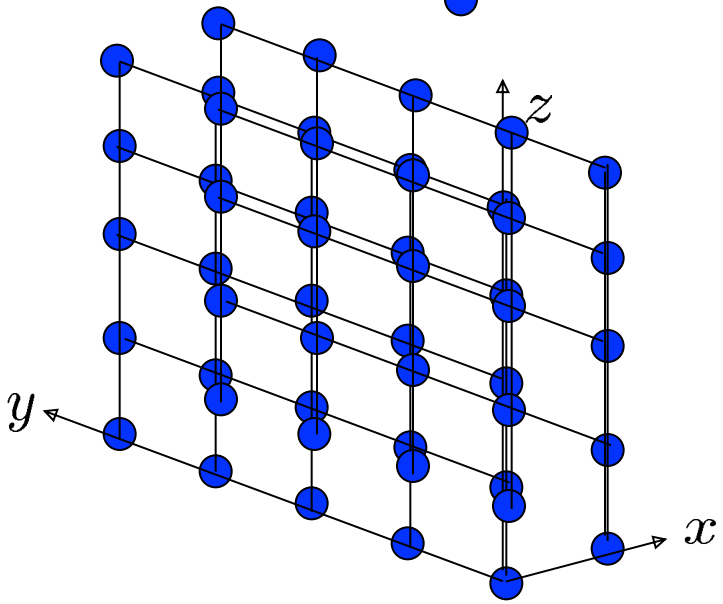
Even number of $\mathbf{X}$s in xy planes

Abelian Invariant Subgroup



Double planes of $\mathbf{Z}$s

$$\mathbf{E}_k = \prod_{i,j} \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1}$$



Double yz planes of $\mathbf{X}$s

$$\mathbf{F}_i = \prod_{j,k} \mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k}$$

$\mathcal{S} = \langle \mathbf{E}_k, \mathbf{F}_i \rangle \subset \mathcal{A}$ $\quad$ $\mathcal{S}$ is abelian

$$\mathbf{SA} = \mathbf{AS},\ \mathbf{A} \in \mathcal{A},\ \mathbf{S} \in \mathcal{S}$$

$\mathcal{S}$ is an abelian invariant subgroup of $\mathcal{A}$

A Painful Construction



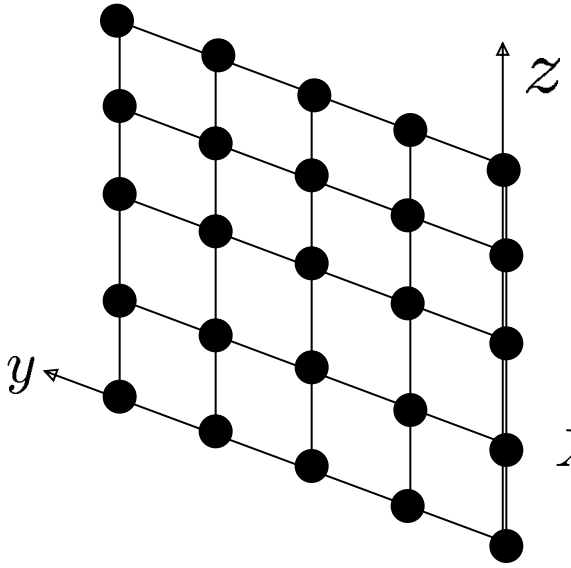
Group elements can be written $\mathcal{SP} = \mathcal{PS}$

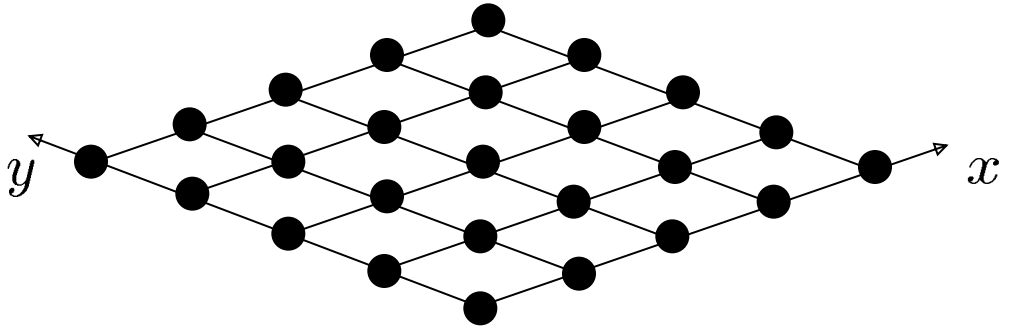
$\mathcal{S}$ is the abelian invariant subalgebra

$\mathcal{P}$ is the Pauli group on $n^3 - 2n + 1$ qubits

The construction for the Pauli group is beyond ugly

But $n^3 - (n^3 - 2n + 1 + 2(n-1)) = 1$, so there must be one encoded qubit

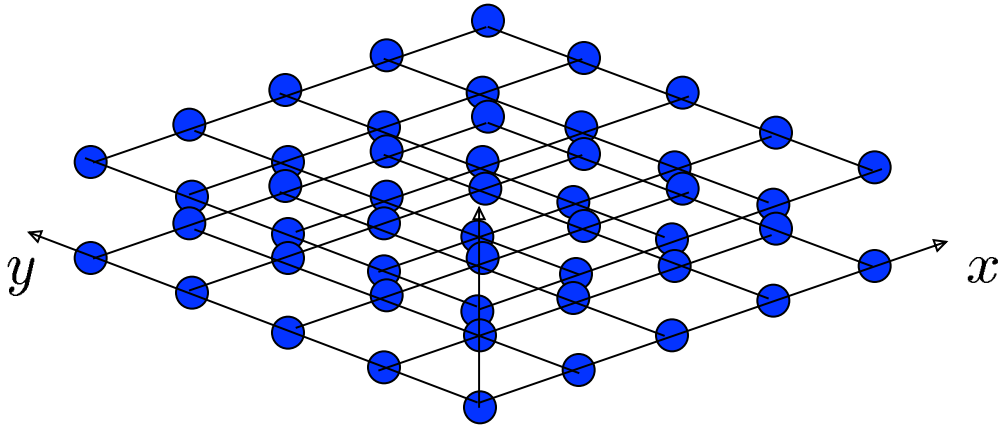




$\bar{Z}$ =Product of n^2 **Z**s in xy plane

$\bar{X}$ =Product of n^2 **X**s in yz plane

The Subsystem Code



Double planes of $\mathbf{Z}$s

$\mathbf{E}_k = \prod_{i,j} \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1}$

Determines if odd or even number of $\mathbf{X}$ errors in double xy plane

Each xy plane has even or odd $\mathbf{X}$ errors

$e,\ e,\ e,\ e,\ e,\ \ldots,\ e \quad \mathbf{E}_k = e$

$error$

$e,\ e,\ o,\ e,\ e,\ \ldots,\ e$

$\mathbf{E}_2 = o \quad \mathbf{E}_3 = o$

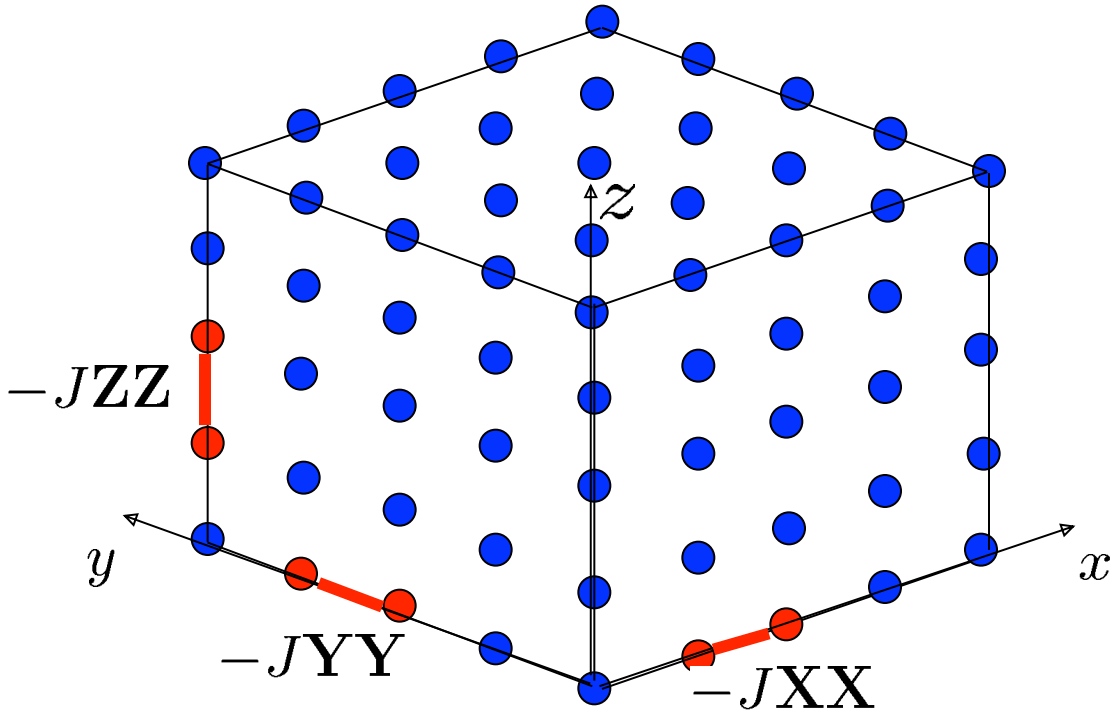
Using $\mathbf{E}_k$ measurements we can fix $(n-1)/2$ odds to evens

If all evens, the $\bar{\mathbf{Z}}$ has been left invariant!

But evens are elements of $\mathcal{P}$!

Same for yz planes

Fin



doubly degenerate energy levels

subsystem code

can write down order parameter for subsystem code

no sign problem

intuition of stringlike excitations